



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Are the stages of an ASP.NET page life cycle similar to those of an application?



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Represent the solution outline to the problem presented by the customer through making case of use diagrams, facing the problem to the customer following rules and protocols of the organization.

Ask your partner:

- What type of content files does an ASP.NET project have?
- What do you need to run an ASP.NET project?

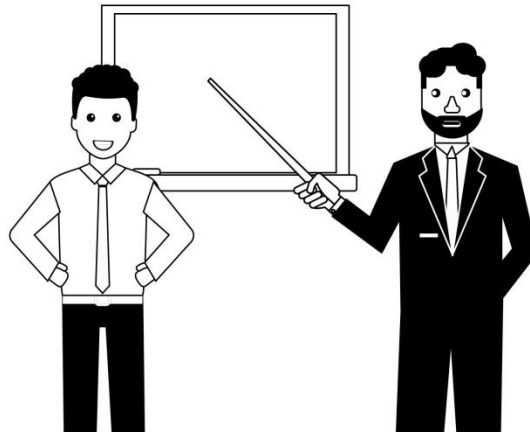
Can Do Statements:

- I can talk the life cycle for an ASP.NET application.
- I can list the stages in the life cycle of an ASP.NET application.
- I can discuss the ASP.NET page life cycle.



Let's Explore!

Sean is taking a tutoring session about ASP.NET applications and pages life cycles.



Instructions: Read the text below.

Tutor: To have a better control of an ASP.NET application, programmers need to understand all the processing phases of an application life cycle.

Sean: What kind of data is processed in a life cycle?

Tutor: Well, you know, a life cycle determines essential steps of an application such as pages processing, instantiation and dynamic compilation.

Sean: But, what are the stages of an application life cycle?

Tutor: Take a look at this diagram, once a user requests to have access to an application resource, a browser sends the request to a server. After this, a unified pipeline gets the information in order to analyze it. In this stage, an Application Manager object is created as well as an object of the class Hosting Environment.

Sean: Does the compilation happen there?

Tutor: That's right; the compilation of all top level items takes place at that point.

Sean: What happens next?

Tutor: Response objects, including Http Context, Http Request and Http Response are created for each request. Then, an Http Application object is set and the request data is processed.

Sean: So, that means that Http Application pipeline does most of the processing work, doesn't it?

Tutor: That's right, once the Http Application takes the request, it starts a series of tasks such as authorization, resolve resolution, state acquisition, handler execution or log request.

Sean: Are the stages of an ASP.NET page life cycle similar to those of an application?

Tutor: They're kind of different; the word "SILVER" can help you remember those phases easily. S refers to "start" in this phase. The page defines properties of request and response. I is for "initialization", there, programmers can set the page controls. L represents "load", the stage in which all control properties are activated; they perform tasks based on the information provided by view state and control state.

Sean: If I'm not wrong, V is for validate, E for event handlers, and R for render...





Let's Practice!

Instructor talks about the life cycle for an ASP.NET application.

Instructions: Read the statement. Select all the correct answers.

1. Which of the following sentences are related to the life cycle for an ASP.NET application?
 - a. Http Application object is set and the request data is processed.
 - b. An Application Manager object is created as well as an object of the class Hosting Environment.
 - c. Internet has been one of the most powerful tool since 2000.
 - d. Http Request and Http Response are created for each request.
 - e. Engineers create many programs every day.



Let's Practice!

Do you know the meaning of these words?

Instructions: Select the correct answer. Read the sentences aloud. Compare your pronunciation with a classmate.

1. An application Manager _____ to control application domains in the hosting environment.
 - a. is assigned
 - b. is read
 - c. is written
2. Hosting Environment is evaluated to get information about an application such functions and services.
 - a. is seen to
 - b. is evaluated at
 - c. is evaluated to
3. View State _____ a method to save information about the value of the page and controls during a session.
 - a. is followed as
 - b. is seen as
 - c. is seen to
4. Handler Execution _____ as a process that gives response to an http request.
 - a. is taken to
 - b. is seen to
 - c. is taken



Extension Activity

Let's design a crossword.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about the life cycle for an ASP.NET application.
- Choose the most important vocabulary related to the life cycle for an ASP.NET application
- Design a crossword with the vocabulary that you chose.
- Share your crossword with a classmate.

Self-Assessment

Yes

No

- I look for information about the life cycle or an ASP.NET application. ☐ ☐
- I design a crossword with vocabulary related to the life cycle for an ASP.NET application. ☐ ☐
- I share my job with a classmate. ☐ ☐



Target Words

- **instantiation** (noun): a process to create a real representation of a class; a concrete representation of a class is an object
e.g. A life cycle determines essential steps of an application such as pages processing, instantiation and dynamic compilation.
- **unified pipeline**(noun): a server processing mode that can run operational steps faster in ASP.NET
e.g. After this, a unified pipeline gets the information in order to analyze it.
- **application manager** (noun): a class that controls application domains in the hosting environment
e.g. In this stage, an Application Manager object is created as well as an object of the class Hosting Environment.
- **hosting environment** (noun): a class that has information about an application such functions and services
e.g. In this stage, an Application Manager object is created as well as an object of the class Hosting Environment.
- **handler execution** (noun): a process that gives response to an http request
e.g. Once the Http Application takes the request, it starts a series of tasks such as authorization, resolve resolution, state acquisition, handler execution or log request.
- **view state** (noun): a method to save information about the value of the page and controls during a session
e.g. They perform tasks based on the information provided by view state and control state.



Learning Points

Be Created and Be Processed to Talk About the ASP.NET Life Cycle

You use the structure **be + verb in past participle** to describe the stages of a process.

You can use the structures **be + created** or **be + processed** to talk about the different steps involved in an ASP.NET page or application cycle.

Remember that you have to conjugate the verb Be according to the subject.

- During this stage, request and response properties **are created**.
- The request **is processed** in a page life cycle.

You can use this structure in affirmative, negative and interrogative statements:

- During this stage, request and response properties **are created**.
- During this stage, request and response properties **aren't created**.
- **Are** request and response properties **created**, during this stage?

Examples

- *In this stage, an Application Manager object is created as well as an object of the class Hosting Environment.*
- *Response objects, including Http Context, Http Request and Http Response are created for each request.*
- *Then, an Http Application object is set and the request data is processed.*



Learning Points

Understanding the Stages of an ASP.NET Page

There are some important steps in a page life cycle. They start when a user wants to have access to a page.

The page life cycle of a ASP.NET page is defined mainly by six fundamental stages that can be memorized using the acronym SILVER:

1. **Start**, in this stage, the page defines properties such as request and response and it can also determinate if it is dealing with a postback to activate the IsPostBack property.
2. **Initialization**, a phase in which programmers can set the page controls and if necessary they can apply a master page and themes.
3. **Load**, control properties are activated; they perform tasks according to the information gathered by view state and control state methods.
4. **Validate**, here the validation method takes places and if the operation is correct, the IsValid property is checked.
5. **Event handlers**, this stage is necessary only when the request is a postback.
6. **Render**, the event when the page runs the render method for all the page controls.

Finally, at the end of the process, the page gets back to the server, and the request and response properties are unloaded.

